Seminar 09 cheat sheet

Linear Programming and the Simplex Method

Optimization Methods
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Forms

- $\max c^\top x$, $Ax \le b$, $x \ge 0$; standard: $\min c^\top x$, $Ax = b$, $x \ge 0$.
- Row $\le$: $+s$; row $\ge$: $-s$; free $x = x^+ - x^-$; $b_i < 0$: row times -1; $\max c^\top x = -\min(-c^\top x)$.
- Outcomes: infeasible, unbounded, optimal.

Corners

- Basis: m independent columns; $x_B = B^{-1}b$, others zero.
- Corner $\Leftrightarrow$ basic feasible solution, $x_B \ge 0$.
- At most $\binom{n}{m}$ bases. An optimum, if any, is at a corner.

Simplex step (max)

- Last row $z - c^\top x = 0$; basis of slacks if $b \ge 0$.
- Enter: $r_j < 0$ (Dantzig: the most negative; Bland: smallest index). None: optimal.
- Leave: smallest b_i/a_{ie} over $a_{ie} > 0$. No positive entry: unbounded.
- Pivot: make the column a unit column, z-row included.
- Zero under a nonbasic variable: another optimum may exist.

Phase I

- Make $b \ge 0$; rows without a basic variable get artificial $a_i \ge 0$.
- Minimize $w = \sum a_i$; price out the basic a_i first.
- $w^* > 0$: infeasible. $w^* = 0$: pivot out the a_i left at zero, restore z, price out, go on.

Duality

- $\max c^\top x$, $Ax \le b$, $x \ge 0 \leftrightarrow \min b^\top y$, $A^\top y \ge c$, $y \ge 0$.
- Row $=$ gives free y_i; free x_j gives row $=$.
- Weak: $c^\top x \le b^\top y$. Strong: equal at the optimum.
- Unbounded primal $\Rightarrow$ infeasible dual.
- Slackness: $y_i(b - Ax)_i = 0$, $x_j(A^\top y - c)_j = 0$.
- $y^\top = c_B^\top B^{-1}$: the z-row under the slacks.

Sensitivity

- $\partial p^*/\partial b_i = y_i^*$ while the basis lasts; $p^*(b)$ is concave, piecewise linear.
- Range of b_i: $x_B + \delta B^{-1}e_i \ge 0$; $B^{-1}e_i$ is the column of s_i.
- Range of c_j: keep the z-row nonnegative.
- Slack row: price zero, but it can still limit a range.

Failures

- Degenerate: a basic variable is zero, a pivot may not move the point.
- Cycling: bases repeat. Bland's rule: smallest index in and out.
- Worst case 2^n corners (Klee–Minty); usually a few times m.
- `linprog`: status 2 infeasible, 3 unbounded; `ineqlin.marginals` are the prices of a min.

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